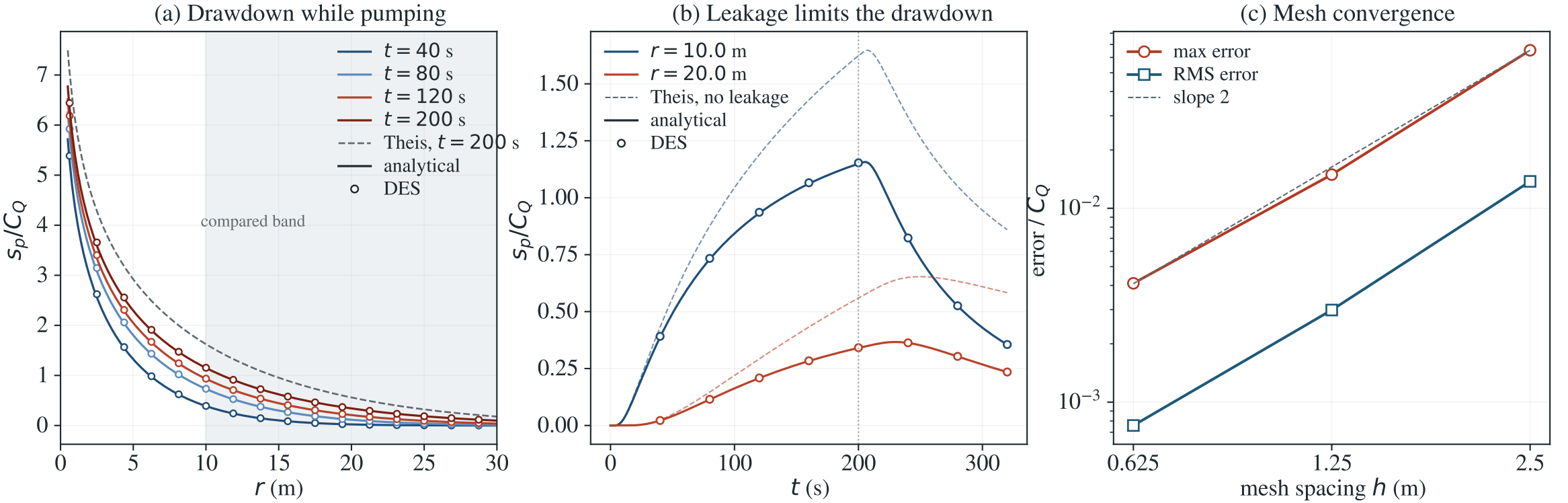

(a) Drawdown while pumping
$t = 40$ s
$t = 80$ s
$t = 120$ s
$t = 200$ s
Theis, $t = 200$ s
analytical
DES
compared band
s_p/C_Q
r (m)
(b) Leakage limits the drawdown
$r = 10.0$ m
$r = 20.0$ m
Theis, no leakage
analytical
DES
s_p/C_Q
t (s)
(c) Mesh convergence
max error
RMS error
slope 2
error / C_Q
mesh spacing h (m)